

# A 900MHz Shielded Bow-tie Antenna System for Ground Penetrating Radar

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**Abstract**—A back-cavity shielded bow-tie antenna system working at 900MHz center frequency for ground-coupled GPR application is investigated numerically and experimentally in this paper. Bow-tie geometrical structure is modified for a compact design and back-cavity assembly. A layer of absorber is employed to overcome the back reflection by omni-directional radiation pattern of a bow-tie antenna in H-plane, thus increasing the SNR and improve the isolation between T and R antennas as well. The designed antenna system is applied to a prototype GPR system. Tested data shows that the back-cavity shielded antenna works satisfactorily in the 900MHz GPR system.

**Keywords**—Ultra-Wideband (UWB), Ground-penetrating radar (GPR), back-cavity, bow-tie

## I. INTRODUCTION

Ground-penetrating radar (GPR) techniques are being applied widely for the detection and identification of buried objects and structures [1]. The working frequency and bandwidth of a GPR system are crucial to the system performance. The higher working frequencies are needed for the better resolution in order to detect small-size targets, while the lower frequencies are preferred for detecting some deep-buried objects [2,3]. Thus, ultra-wideband (UWB) GPR system that transmits short time impulse has the advantages of working from low to high frequencies. UWB transmitter and receiver antennas are desired to have flat and high directivity gain, narrow beam, low side-lobe and input-reflection levels over the operational frequency band in order to reach the maximum dynamic range, best focused illumination area, lowest T/R antenna coupling, reduced ringing, and uniformly shaped impulse radiation [4,5,6]. For a UWB GPR device, portability is also currently becoming an issue. Considering factors such as compact geometrical size, instantaneous bandwidth, waveform fidelity, radiation efficiency and convenient engineering manufacturing, bow-tie antenna is often selected for ground-coupled GPR applications.

## II. ANTENNA DESIGN

In practical engineering applications, bow-tie antennas as transfiguration of infinite bi-conical antennas are well accepted not only because they have broadband characteristics but also easy to make because of plane surface structure [7].

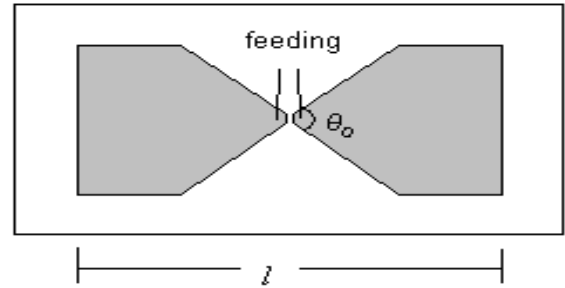


Figure 1. bow-tie structure

Figure 1 shows the structure of the proposed bow-tie antenna. The characteristic impedance of a bow-tie antenna can be calculated by:

$$Z_c = 120 \ln \cot \frac{\theta_0}{4}, \quad (1)$$

where  $\theta_0$  is the opening angle of each side. The bandwidth increases when  $\theta_0$  increases.  $\theta_0$  is usually between 60° and 90° to obtain wideband performance and compact geometrical structure. The length of antenna is critical to the low frequency performance. The longer the length is, the better the lower frequency performance. The length of bow-tie is determined by [8]:

$$l = \frac{1}{2} \lambda_0 \times \frac{1}{\sqrt{\epsilon_{eff}}}, \quad (2)$$

which  $\lambda_0$  is the wavelength corresponding to the expected lowest working frequency. In this paper we expect the system working around a center frequency of 900 MHz and lowest frequency limit is set to 400 MHz.  $\epsilon_{eff}$  is the effective dielectric constant relative to bow-tie width  $w$ , substrate dielectric constant  $\epsilon_r$  and substrate thickness  $d$ , which can be calculated by following expression [8]:

$$\epsilon_{eff} = \frac{\epsilon_r + 1}{2} + \frac{\epsilon_r - 1}{2} \left( 1 + 10 \frac{d}{w} \right)^{-0.555}, \quad (3)$$

### A. Edge-cutting of Bow-tie

As a common truth for the bow-tie antenna, the performance at low frequency is basically determined by the bow-tie length, while the bandwidth is affected by the opening angle  $\theta_0$ . Under this consideration, a parallel-cutting on the

bow-tie edges is employed to minimize the geometry dimension of the bow-tie antenna, as figure 1 shown. Theoretically, edge-cutting will cause the current reflection on the edge and also change the constant impedance characteristics of bow-tie. Simulations are presented to illustrate the effects of different edge-cuttings on antenna performance.

Figure 2(a) to 2(d) present the geometry structures of bow-tie with different cutting levels. Figure 3(a) to 3(d) accordingly show the S11 simulation results. The discussion comes to a brief conclusion that, when less than  $l/2$  of the bow-tie width is cut off, the center frequency shifts not significantly. As figure 3(a), 3(b), 3(c) shown, the center-frequency of bow-tie B shifts about 8% higher than bow-tie A, and the center-frequency of bow-tie C shifts about 2% higher than bow-tie B. In the meanwhile, the bandwidth of bow-tie B increases about 21% than bow-tie A, and the bandwidth of bow-tie C increases about 5% than bow-tie B as well. Moreover, the return loss is getting lower by increasing the cut-off area. However, when the cut-off width is more than  $w/2$ , as figure 2(d) shown, the S11 parameter of bow-tie D is getting worse as to bow-tie C, and the center-frequency shifts evidently (about 28% higher than bow-tie A). Therefore, the bow-tie C is considered as the appropriate geometry structure to be applied in the GPR system.

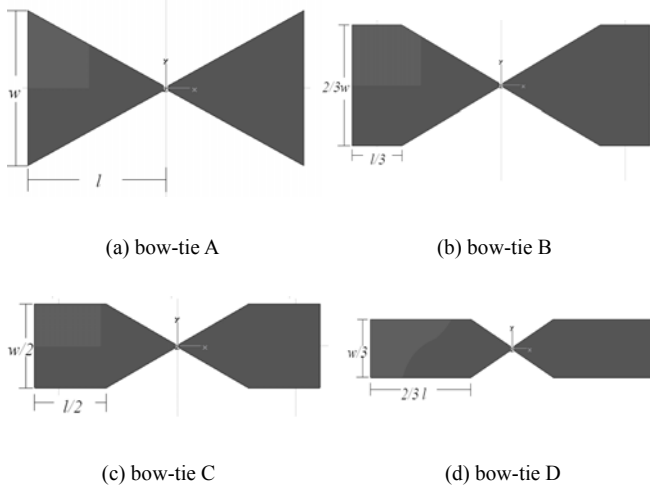
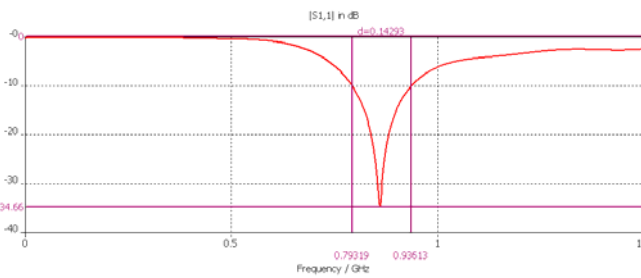
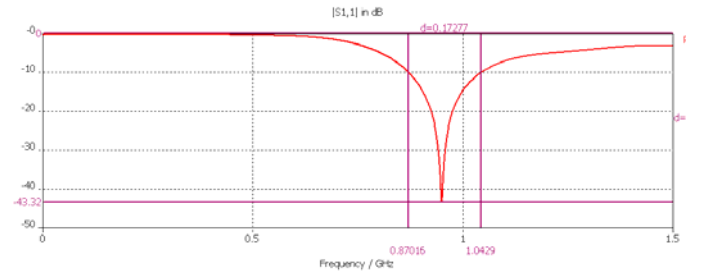


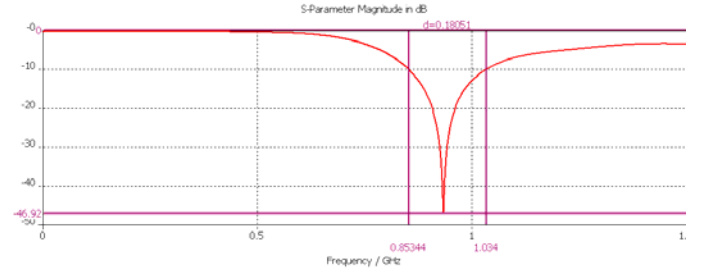
Figure 2. bow-tie with different edge-cutting levels



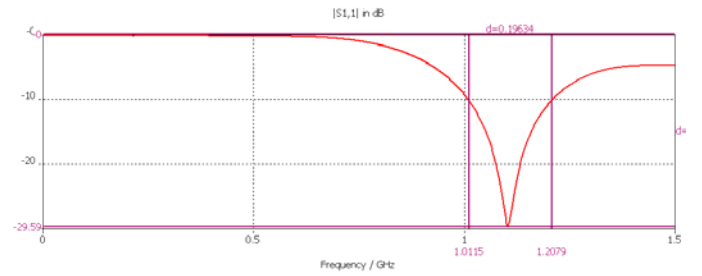
(a) S11 of bow-tie A



(b) S11 of bow-tie B



(c) S11 of bow-tie C



(d) S11 of bow-tie D

Figure 3. S11 parameters of different bow-tie structures

## B. Back-Cavity

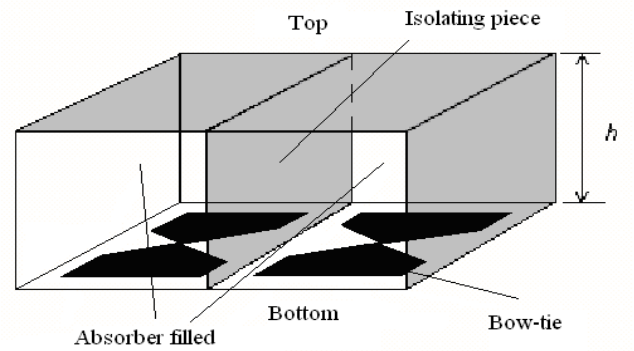


Figure 4. back-cavity bow-tie with absorber

Radiation pattern of bow-tie antenna is omni-directional in H-plane since we desire single side radiating which is facing to ground for improved radiation efficiency and avoid noise from upper side. A metal cavity structure shown in figure 4 is

designed for a bow-tie antenna to prevent external noise as well as to reduce unnecessary air radiation<sup>[9]</sup>.

Effective back-cavity design could reinforce the radiation towards ground surface as well as shielding the interference signals and suppressing the back-radiation. The crucial factor to determine the antenna performance is the dimension of the back-cavity. The cavity acts as a wideband reflector to the signal radiated from the back of the antenna. Multiple reflections will deteriorate the GPR signal if multiple reflections are not absorbed. Therefore, absorbing materials must be inserted in the cavity, which will be discussed in section B. Design of the cavity can be done by numerical simulation.

As common knowledge any cavity with certain dimension means a narrowband system, therefore we need to design this back-cavity system working at the center frequency of the GPR system. As shown in Figure 2, bow-tie antenna with shield can be considered as a two element antenna array. The entire radiating field to the ground can be described as :

$$E = E_1(1 - e^{j2kh}) \quad (4)$$

which  $E_1$  is the radiating field amplitude of each bow-tie element; and  $k = 2\pi / \lambda'$  is the wave number,  $\lambda'$  is the antenna wavelength of center frequency in the absorbing material;  $h$  is the height of cavity. It is apparent the  $E = 2E_1$ , which means energy radiating to the ground reaches maximum when  $h$  is  $\lambda'/4$ . As shown in Figure 2, bow-tie antenna is placed on the bottom side of the open cavity, and the side surfaces of the cavity has minor effects since they are vertical to the  $E$  field of the antenna radiation.

To reduce the size of the antenna, the space between the edge of the antenna and four sides of the shield must be minimized. On the other hand, the space must be large enough to prevent the four sides from affecting antenna performance<sup>[9]</sup>.

### C. Absorbing Material

As discussed above, undesired multiple reflection exists between antenna and back shield, which will deteriorate the GPR performance<sup>[9]</sup>. Thus, preventing multiplexed reflection by loading absorbing material inside the cavity must be considered.

The essential idea of absorbing material is resistive loading the antenna. It has distributing resistive property, which will reduce the reflection of upper side of antenna in the cavity. Commercial absorbing material with a thickness of 57.4mm was used. The absorbing material has a gradient resistivity distribution to reduce reflection. The material can attenuate back radiation by more than 30 dB in the range of 570 MHz to 2400 MHz. Figure 5 shows the receiving signal waveform comparison of the shielded bow-tie antenna with and

without absorber filling. It proves the absorber effects on suppressing the multi-reflection or ringing of bow-tie antenna.

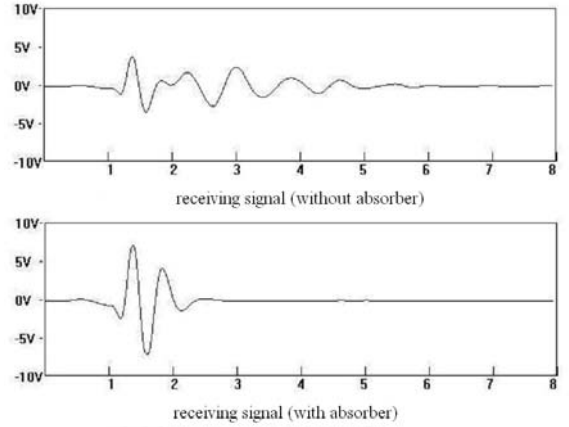


Figure 5. absorber effects (field waveforms)

### D. Wide band Impedance Matching

Impedance matching is one of the most critical issues in UWB antenna design. Unfortunately it is difficult to make it exactly matched in a wide band. There are two critical parts we need to match the impedance, one is between transmitter and antenna feeding port, the other is between antenna aperture and air or ground, depends on applications. Ground surface and material present a load to the antenna. For the ground-coupled GPR system, the characteristic impedance can be calculated by expression (1) when antenna is in air, which is around 158 ohms when  $\theta_o=60^\circ$ . However, the characteristic impedance of the feeding line is usually 50 Ohms and the impedance of transmitter is even much lower than 50 Ohms. Thus there is an obvious mismatch existing between transmitter/coaxial connector and antenna feeding port. To solve this problem, wideband matching scheme is used in antenna feeding structure design as shown in Figure 6.

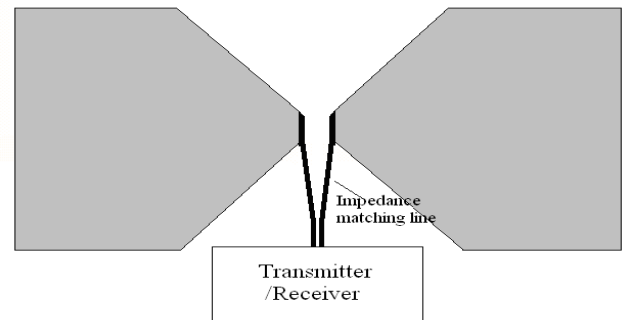


Figure 6. impedance matching line structure

A pair of gradually transforming line is designed to form a wideband matching circuit. The spacing between each transmission line section can be designed and numerically simulated to optimize the antenna performance by software.

A balun transformer can be used to convert the balanced input of the antenna to the un-balanced output of the GPR transmitter. Careful adjustment of the length and spacing of

each section can optimize the matching during fine tuning of the antenna performance. The bow-tie antenna measures 170mm×60 mm,  $\theta_o=60^\circ$ , with a FR4 substrate of  $\epsilon_r = 4.6$ , and  $d=1.67\text{cm}$ . The dimension of the shielding box is 180mm×180mm×80mm, made of 1mm-thick aluminum sheet.

### III. ANTENNA PERFORMANCE

Using the antenna designed above, a 900 MHz GPR system is setup. Transmitting signal is differential pulse with 1ns width. The GPR system is battery operated and data is transmitted to a computer via Ethernet. Figure 6 is the implemented shielded antenna system. The transmitting and the receiving circuits are integrated inside the shielding box. Figure 7 presents the proposed 900MHz GPR prototype (the left one in the picture). A 400MHz GPR prototype<sup>[6]</sup> which has been accomplished in previous work is also presented in the same picture as a contrast (the right one in the picture). Distance Measuring Instruments (DMI) is implemented as well to the 900MHz GPR prototype for advanced engineering application.



Figure 6. Implemented shielded antenna system



Figure 7. GPR prototypes

Figure 8 presents a measured data when the GPR is placed over a soil with dielectric constant of 5. It is clearly seen that the multiple reflection and ringing is minimized.

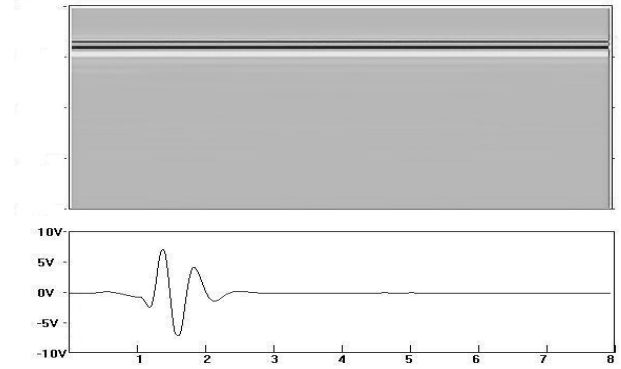
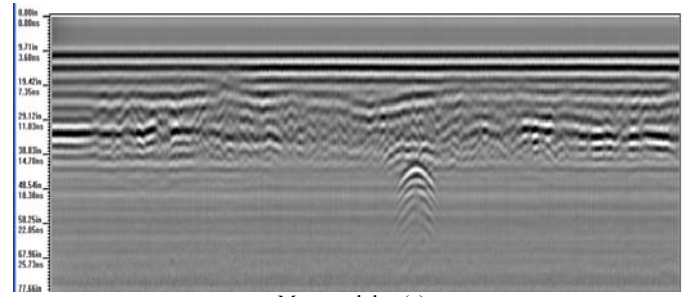
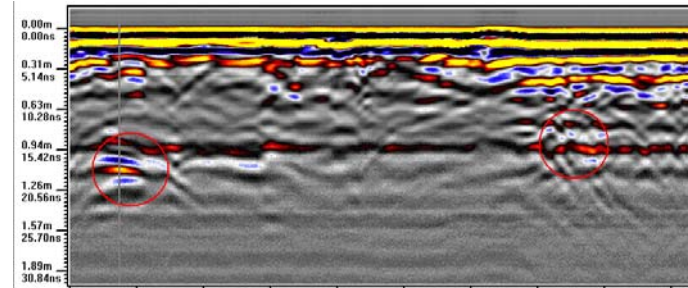


Figure 8. Measured signal over a soil

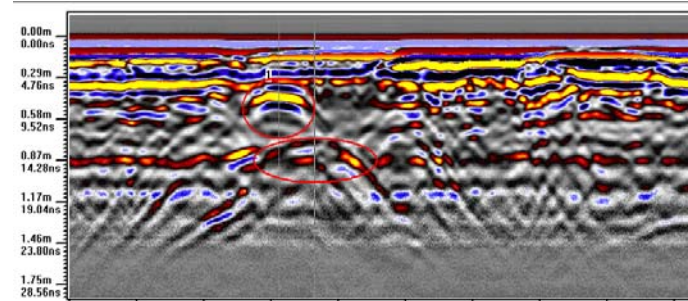
Figure 9 presents several examples of the GPR measurement. Figure 9(a) is the measurement over a concrete water pipe at a depth of 0.6m. Figure 9(b) is the measurement of pipes which are under asphalt pavement (circled areas are concrete pipes, which are about 0.65m and 0.93m in depth). Figure 9(c) is a similar measurement of pipes under asphalt pavement. The pipes are about 0.3m and 0.87m in depth.



Measured data(a)



Measured data(b)



Measured data(c)

Figure 9. GPR Measured data

#### IV. CONCLUSIONS

A compact GPR antenna system is designed and implemented. The practical design guidance is discussed. The designed shielded bow-tie antenna is used in a GPR system working at 900MHz center frequency. With properly shielding and absorbing design, this antenna system has less ringing, better resolution and a compact structure. Measured data shows that the bow-tie antenna practically performs as desired.

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